

Selected Exercises 1.1 Find the fixed points of the logistic map, the standard map and the Hénon-Heiles system from Sect. 1.1.1 and discuss their stability. 1.2 Find period-2 orbits of the standard map. Hint: solve $x_{n+2} = x_n$. 1.3 For which parameter values is the Hénon map, $x_{n+1} = 1 - ax_n^2$, $y_{n+1} = x_n$, a conservative system? 1.4 Prove that the Hénon-Heiles system (1.6) is conservative. Derive a condition for the parameters of the Lorenz-63 model (1.5) which has to be fulfilled such that this system is dissipative. 1.5 In a computer define and evolve the following dynamical system $\dot{x} = x - yz$, $\dot{y} = y + xz$, $\dot{z} = -z + xy$ (by default $\sigma = 0.119$, $\rho = 0.1$, $\beta = 0.9$ and initial condition $(3, 3, 3)$) called Gissinger system [10], which models chaotic reversals of Earth's magnetic field. Animate the evolution of the three dimensional trajectory for various values of σ . Hint: writing your own ODE solver is probably too much; instead pick an existing one from your programming language of choice. You may compare with the solution you'd get by using `DynamicalSystems.jl` and Code 1.1. 1.6 Evolve a grid of initial conditions of the standard map, (1.4), each for 1000 steps. Scatter-plot all trajectories of all initial conditions with different colors.

Repeat for various k values in different plots. These plots will highlight that the standard map is a system with mixed state space, which we will discuss more in Sect. 8.2.2. 1.7 Show that for Hamiltonian systems the Hamiltonian is a conserved quantity, $\dot{H} = 0$. Apply this knowledge to the single pendulum: $\dot{\theta} = v$, $\dot{v} = -p \sin(\theta)$. Find its conserved quantity. Simulate trajectories and confirm that they conserve this quantity. 1.8 Write your own code that can calculate a Poincaré section at a predefined (hardcoded) hyperplane corresponding the first variable of a trajectory crossing some constant number (e.g., zero). Use linear interpolation between the trajectory points that “sandwich” the hyperplane to generate the section. Apply your code to, e.g., the Lorenz-63 or Gissinger models. Compare with using the function `poincaresos` from `DynamicalSystems.jl` and confirm your code works as expected. 1.9 Extend the above code to be able to work with arbitrary hyperplanes of arbitrary orientation. Provide sections for, e.g., the Lorenz-63 system for the planes: $x = y$, $y = z$, $z = x$. Hint: what is the mathematical equation that defines a hyperplane in a D -dimensional Euclidean space? 1.10 Create an animation of a Poincaré section that scans a trajectory. Choose a hyperplane of constant orientation, and animate how the section looks as the value that defines the hyperplane changes. For example, if the hyperplane is defined as “when the x coordinate crosses value v ”, then animate the Poincaré section for changing v . Apply this animation to the Henon-Héiles system for different trajectories. 1.11 Show that the Fermi-Ulam map $u_{n+1} = -u_n + \sin n$, $n+1 = n + 2 - u_n + \sin n$

mod 2 is conservative. Evolve a grid of initial conditions and scatter-plot them with different color to get a picture similar to the state space of the standard map, Fig. 1.1. Repeat this for various β values. Then, analytically find the fixed points of this system and discuss their stability (Hint: consider $\beta = 0$ or $\beta = 1$). Does what you find make sense with respect to the figures you produced? 1.12 Determine the fixed points of the Lorenz-63 system (1.5). Perform a linear stability analysis of the fixed point $(0, 0, 0)$ of the Lorenz-63 system (1.5) for $\sigma, \rho, \beta > 0$ using the Routh-Hurwitz criterion which states: All roots of the characteristic polynomial of the Jacobian matrix J at the fixed point $(0, 0, 0)$ have negative real parts if and only if the following conditions are satisfied: (i) $a_0 > 0$, (ii) $a_1 > 0$, (iii) $a_2 > 0$, (iv) $a_3 > 0$, (v) $a_0 a_1 a_2 a_3 > 0$.

with the basins. Hint: evolve a set of initial values located close to the fixed point on the unstable/stable subspace given by the eigenvector corresponding to the unstable/stable eigenvalue. The unstable/stable scenario should be evolved forward/backward in time. 1.17 The SIR model [11] (also known as Kermack-McKendrick model [12]) describes the evolution of an epidemic $\dot{S} = -\beta SI$, $\dot{I} = \beta SI - \gamma I$, $\dot{R} = \gamma I$ where S , I , and R are the numbers of susceptible, infected and recovered individuals, respectively. β is the infection rate, and γ is the rate at which individuals recover. Show that the total number $N = S + I + R$ is a conserved quantity. Implement the model and investigate its dynamics for $\beta = 0.5$ and $\gamma = 0.05$ for initial conditions $(S, I, R) = (999, 1, 0)$ and $(S, I, R) = (990, 10, 0)$ and compare the solutions. Then, simulate a lockdown by reducing the infection rate from $\beta = 0.5$ to $\beta = 0.1$ during a time interval $[10, 30]$. 1.18 Consider the system $\dot{v} = \sin v$, $\dot{\phi} = \cos v + v^2/v$.

Show that it has a conserved quantity and identify it. Evolve and plot a couple of initial conditions and confirm that they follow contour lines of the conserved quantity you found. 1.19 We've mentioned that dissipative systems can have co-existing attractors, just like shown in Fig. 1.3, and just like the exercise with the Duffing oscillator. A very interesting case is the Lorenz-84 system: $\dot{x} = y^2 - z^2 - ax + aF$, $\dot{y} = xy - y - bxz + G$, $\dot{z} = bxy + xz - z$ (1.12) with $a = 0.25$, $b = 4.0$, $F = 6.886$, $G = 1.337$. For these parameters the state space has an attracting fixed point, an attracting periodic orbit (limit cycle) and a chaotic attractor, all existing at the same time! Find all three of them! Hint: since you do not have the tools yet to distinguish the different kinds of motion via numerical process, simply plot random initial conditions until you find three where each converges to each of the three attractors.

First document

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In this document some extra packages and parameters were added. There is an encoding package and pagesize and fontsize parameters.